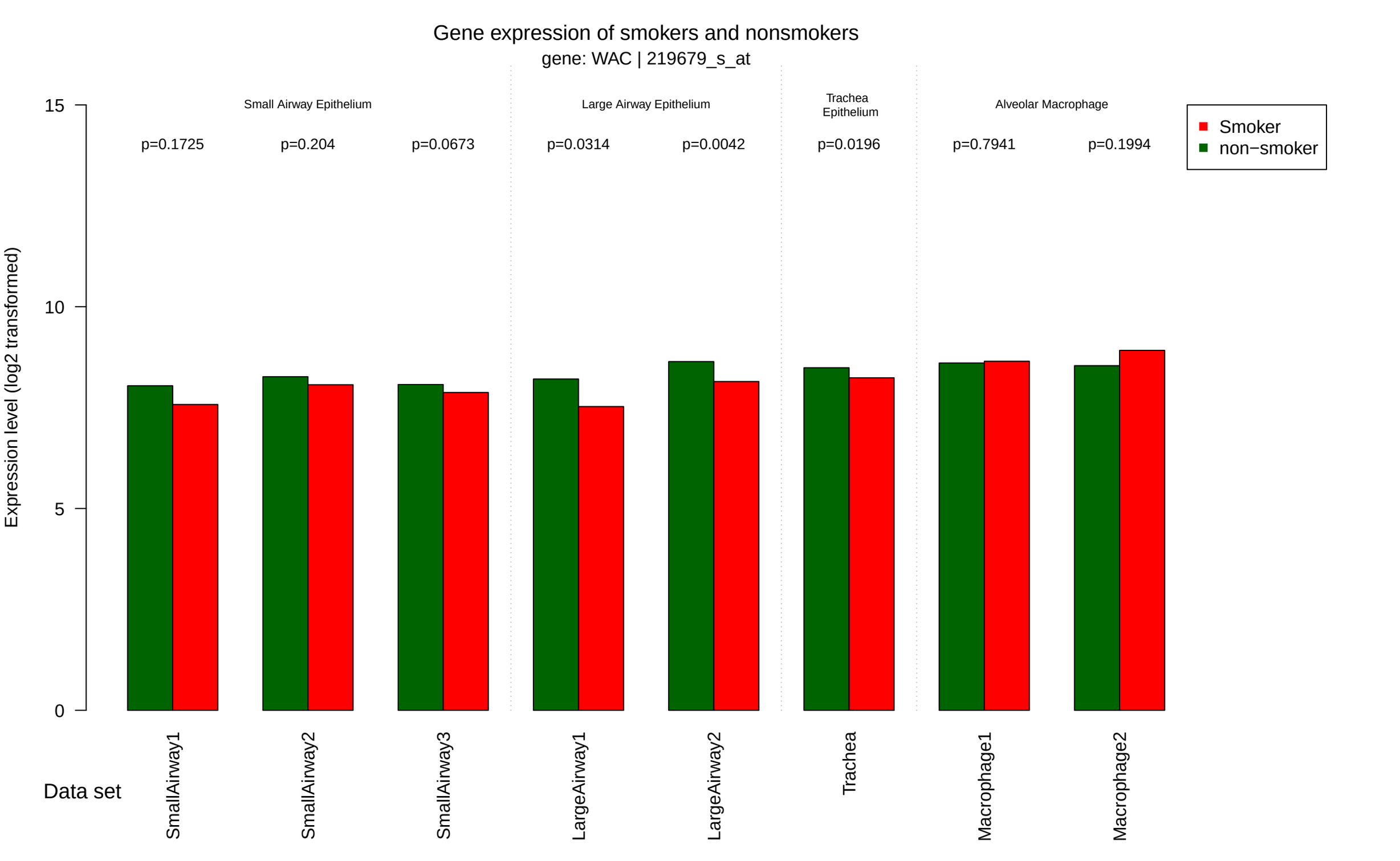
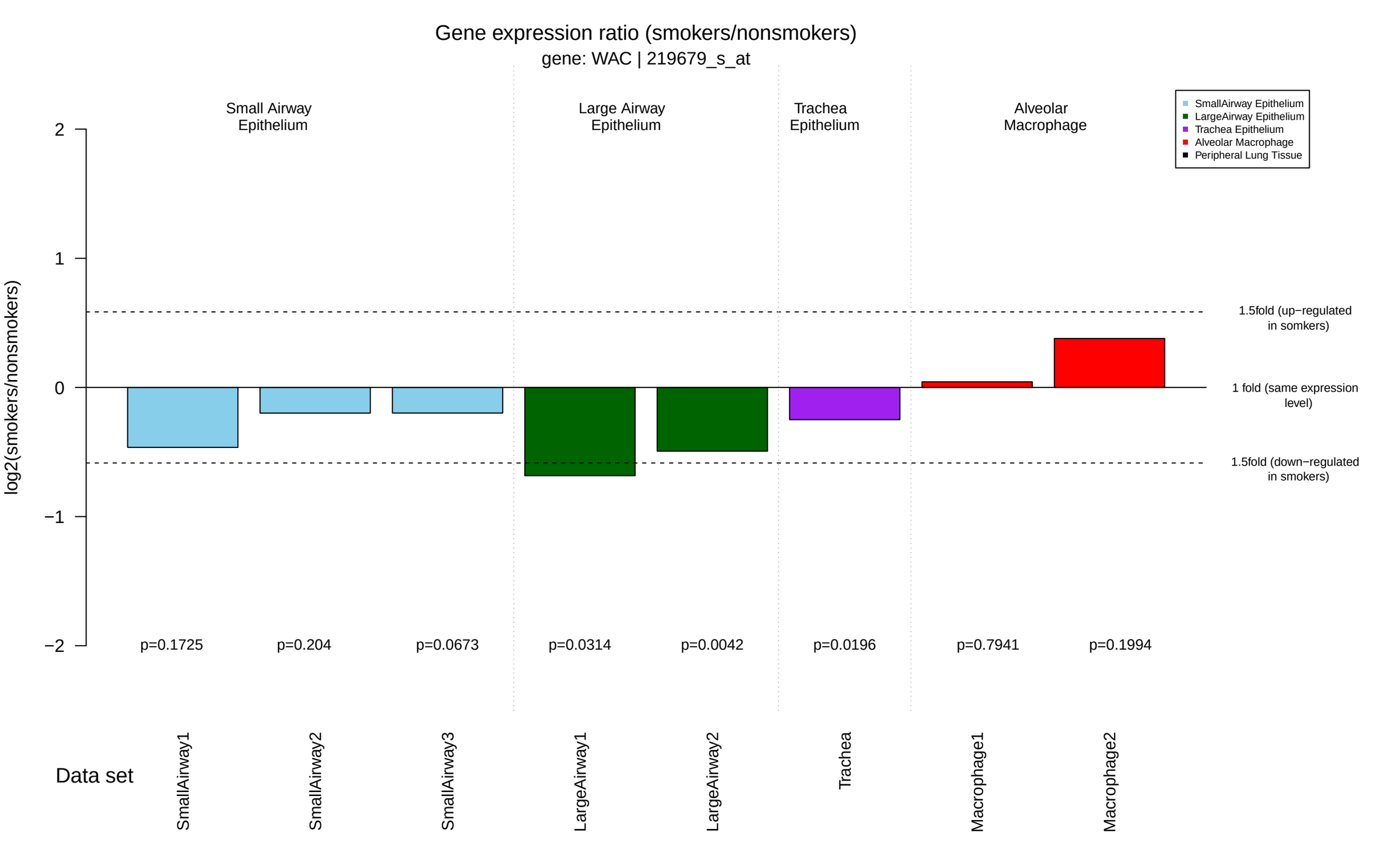
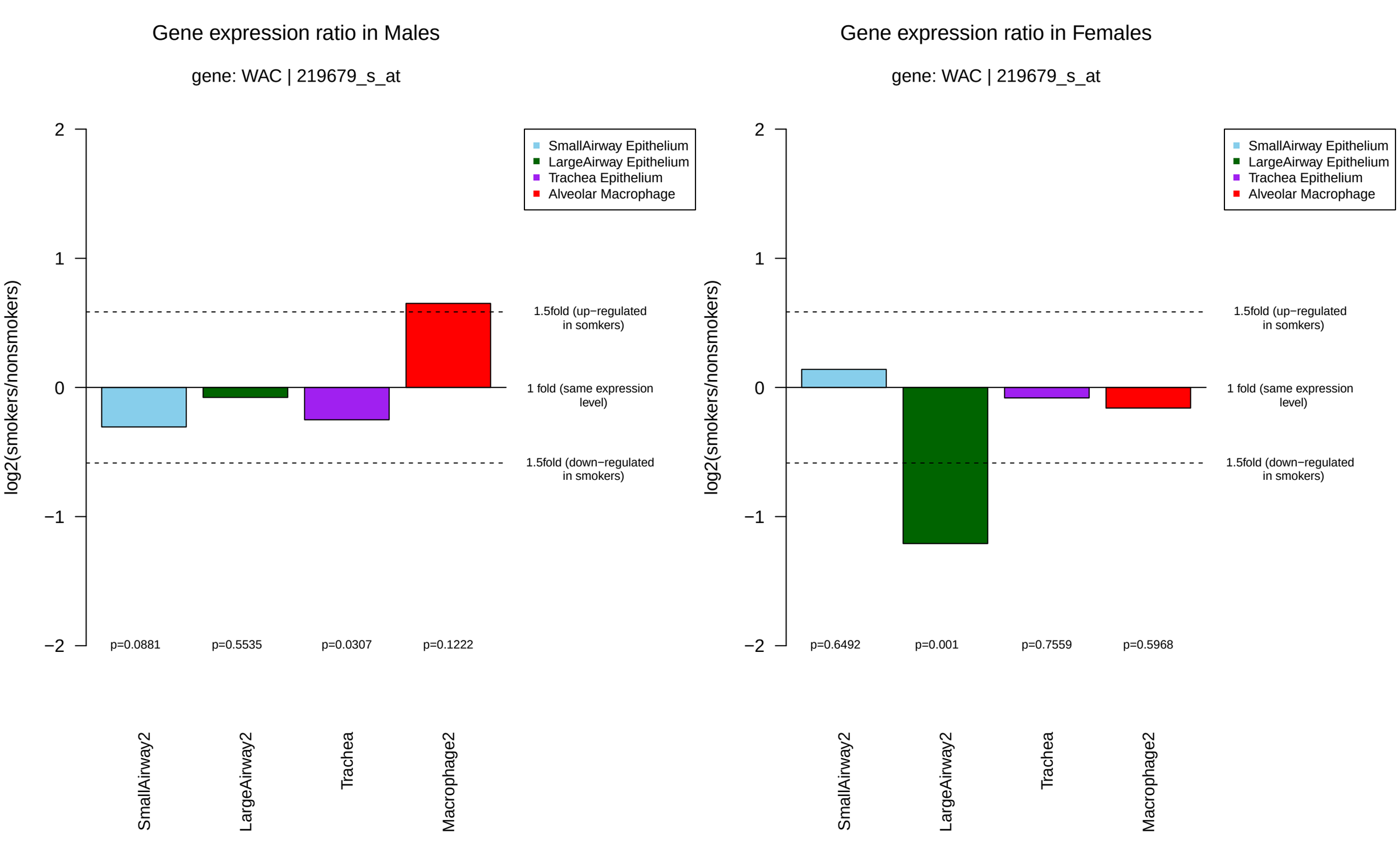
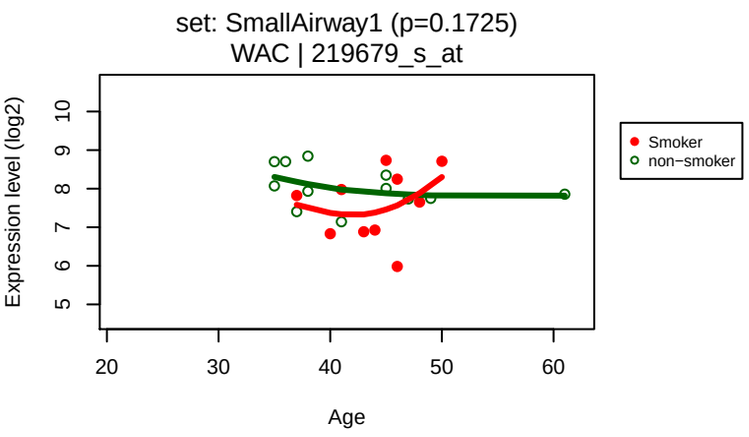
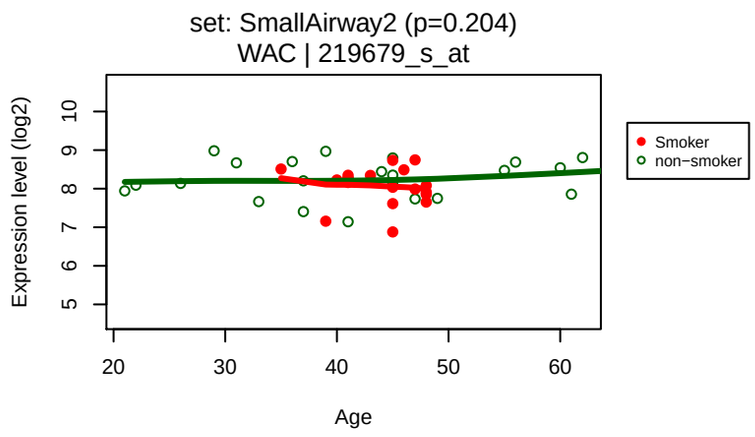
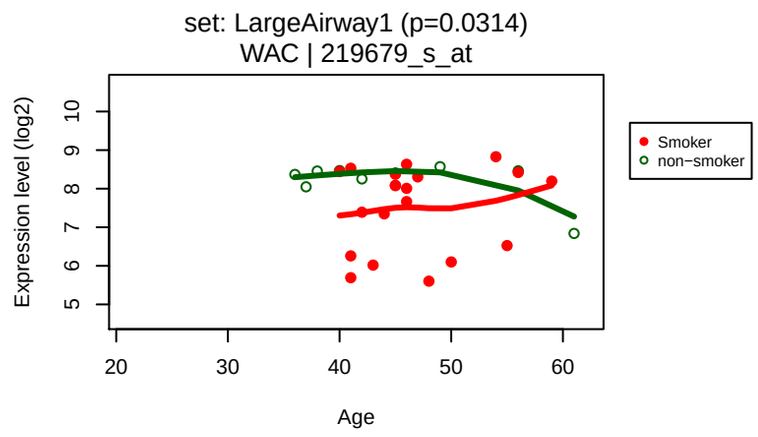

Gene expression of smokers and nonsmokers
gene: WAC | 219679_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1725
p=0.204
p=0.0673
p=0.0314
p=0.0042
p=0.0196
p=0.7941
p=0.1994
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

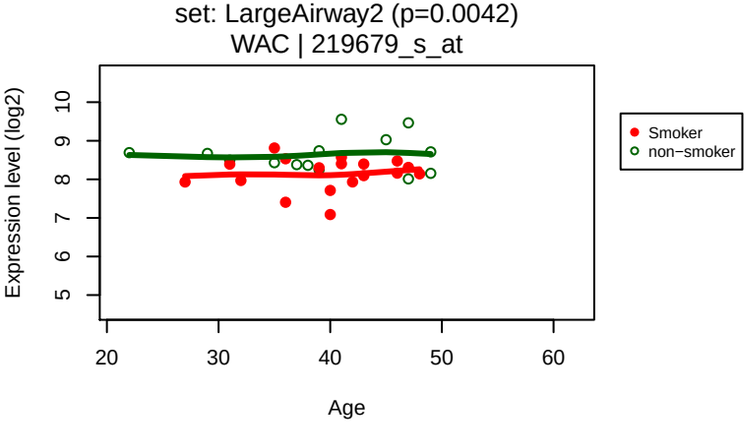

Gene expression ratio (smokers/nonsmokers)
gene: WAC | 219679_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1725
p=0.204
p=0.0673
p=0.0314
p=0.0042
p=0.0196
p=0.7941
p=0.1994
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

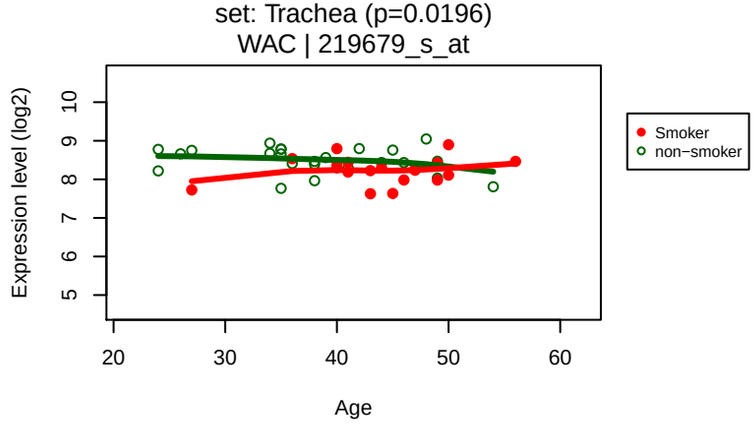

Gene expression ratio in Males
gene: WAC | 219679_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0881
p=0.5535
p=0.0307
p=0.1222
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAC | 219679_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6492
p=0.001
p=0.7559
p=0.5968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

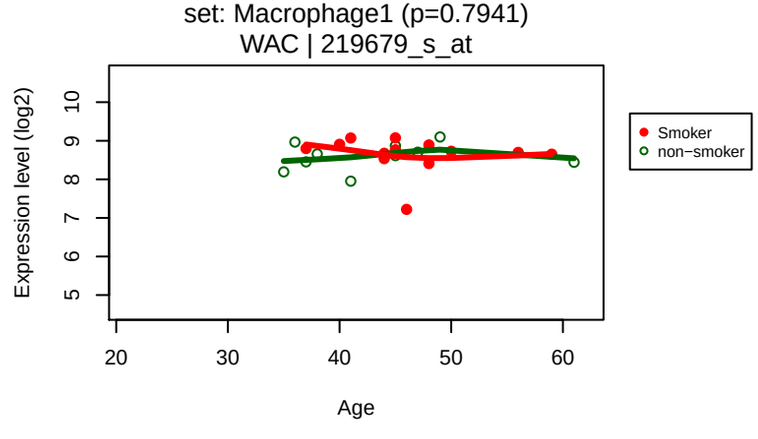

set: SmallAirway1 (p=0.1725)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker

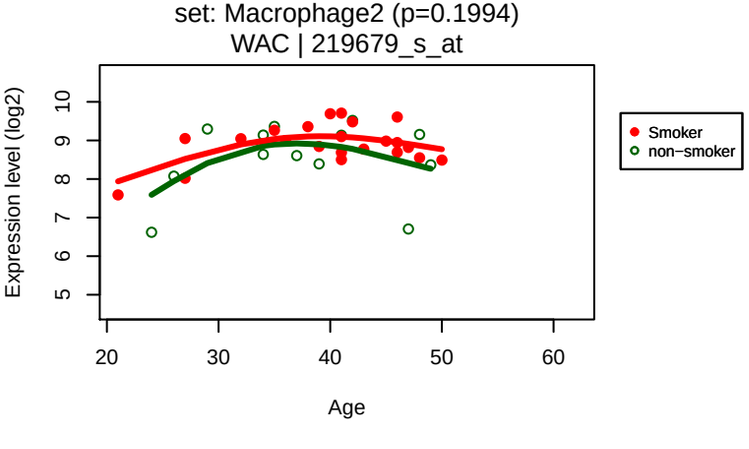

set: SmallAirway2 (p=0.204)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker

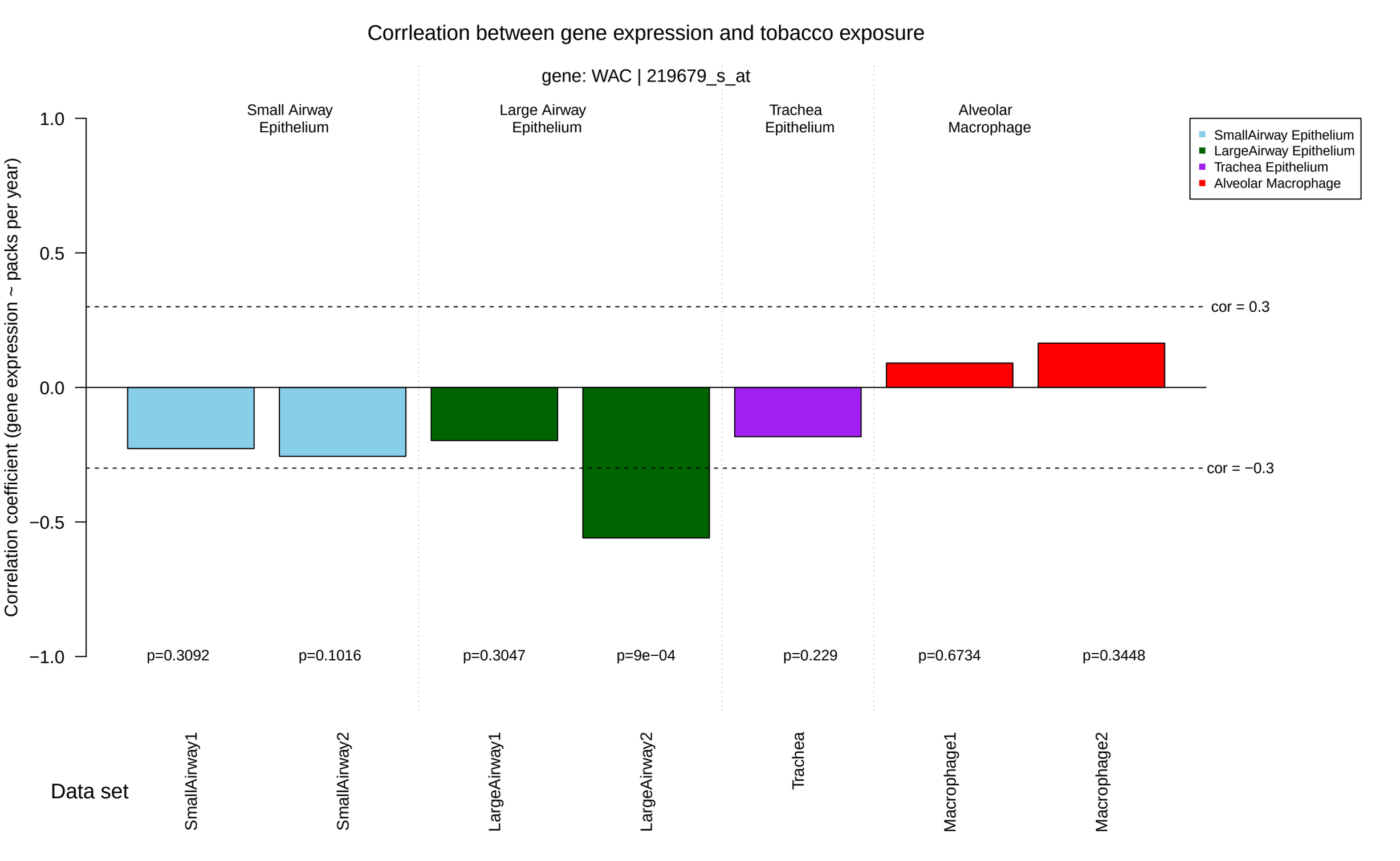

set: LargeAirway1 (p=0.0314)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0042)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0196)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7941)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1994)
WAC | 219679_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAC | 219679_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3092
p=0.1016
p=0.3047
p=9e−04
p=0.229
p=0.6734
p=0.3448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2